Problem Set: The Interpretation of Propositional Formulas

Goldrei §2.3, with emphasis on Theorem 2.2

Throughline. Theorem 2.2 has two halves. Uniqueness says a truth assignment has no freedom beyond its values on the propositional variables; existence says that on the variables the freedom is total. Much of this sheet is about knowing which half you are using at any given moment — and recognizing the situations in which neither is available.

Ground rules

- Notation and conventions follow Goldrei, Chapter 2: formulas are fully bracketed; the *length* of a formula is the number of occurrences of connectives in it; for a function $v\colon P \to \{\mathrm{T}, \mathrm{F}\}$, $\bar{v}$ denotes the unique truth assignment extending v given by Theorem 2.2. As in the book, "all truth assignments" abbreviates "all truth assignments defined on a set of propositional variables including those appearing in the formulas at hand."
- You may use without proof: Theorem 2.2, anything established in the book up to the end of §2.3, and any earlier problem on this sheet.
- **Refutation discipline.** To refute a claim of the form "for all formulas $\varphi, \psi, \ldots$" you must produce a fully concrete instance: (i) name a specific formula for each metavariable; (ii) verify that your formulas satisfy *every* hypothesis of the claim; (iii) exhibit a specific truth assignment, by listing its values on the relevant propositional variables, and compute the truth values needed to show that the conclusion fails. An explanation of why a counterexample ought to exist is not a counterexample.
- Starred problems are harder.

<u>**Theorem 2.2**</u> Let P be a set of propositional variables, let S be the set of connectives $\{\neg, \wedge, \vee, \to, \leftrightarrow\}$ and let $v\colon P \to \{\mathrm{T}, \mathrm{F}\}$ be a function. Then there is a unique truth assignment $\bar{v} : \mathrm{Form}(P, S) \to \{\mathrm{T}, \mathrm{F}\}$ such that $\bar{v}(p) = v(p)$ for all $p \in P$.

Part I — Reading Theorem 2.2

Problem 1 (Reading the theorem precisely)**.** Let P be a set of propositional variables and $S = \{\neg, \wedge, \vee, \to, \leftrightarrow\}$. For each statement below, decide whether it is true or false. Justify each verdict briefly; for false statements, follow the refutation discipline.

(a) For every function $v\colon P \to \{\mathrm{T}, \mathrm{F}\}$ there is at least one truth assignment w with $w(p) = v(p)$ for all $p \in P$.

Response: Let $v\colon P \to \{\mathrm{T}, \mathrm{F}\}$ be arbitrary. By the existence half of Theorem 2.2, there is a truth assignment $\bar{v} : \mathrm{Form}(P, S) \to \{\mathrm{T}, \mathrm{F}\}$ such that $\bar{v}(p) = v(p)$ for all $p \in P$; take $w = \bar{v}$. So (a) is true.

(b) For every function $v\colon P \to \{\mathrm{T}, \mathrm{F}\}$ there is at most one truth assignment w with $w(p) = v(p)$ for all $p \in P$.

Response: Let $v\colon P \to \{\mathrm{T}, \mathrm{F}\}$ be arbitrary. Suppose u and w are both truth assignments with $u(p) = v(p)$ and $w(p) = v(p)$ for all $p \in P$. Then $u(p) = w(p)$ for all $p \in P$ so by the uniqueness half of Theorem 2.2 $u = w$. So (b) is true.

(c) For every function $v\colon P \to \{\mathrm{T},\mathrm{F}\}$ there is exactly one function $w\colon \mathrm{Form}(P,S) \to \{\mathrm{T},\mathrm{F}\}$ with $w(p) = v(p)$ for all $p \in P$.

Response: Let $P = \{p\}$ and let $v(p) = \mathrm{T}$. By Theorem 2.2 there is a truth assignment $\overline{v}$ with $\overline{v}(p) = v(p) = \mathrm{T}$. Define $w\colon \mathrm{Form}(P,S) \to \{\mathrm{T},\mathrm{F}\}$ by: $w(p) = \mathrm{T}$, $w(\neg p) = \mathrm{T}$, and $w(\varphi) = \overline{v}(\varphi)$ for every other formula φ. Let $\phi = \neg p$. Since $\overline{v}$ respects the truth table for $\neg$ and $\overline{v}(p) = \mathrm{T}$, we get $\overline{v}(\phi) = \mathrm{F}$, while $w(\phi) = \mathrm{T}$ by definition. So $w(\phi) \neq \overline{v}(\phi)$. Thus $\overline{v}$ and w are two distinct functions $\mathrm{Form}(P,S) \to \{\mathrm{T},\mathrm{F}\}$ agreeing with v on P, so "at most one" fails. So (c) is false.

(d) For every truth assignment $w\colon \mathrm{Form}(P,S) \to \{\mathrm{T},\mathrm{F}\}$ there is exactly one function $v\colon P \to \{\mathrm{T},\mathrm{F}\}$ such that $w = \overline{v}$.

Response: Let w be arbitrary. Define $v(p) = w(p)$ for all $p \in P$. By Theorem 2.2 $\overline{v}(p) = v(p)$ for all $p \in P$. So $\overline{v}(p) = w(p)$ for all $p \in P$. So by the uniqueness half of Theorem 2.2, $w = \overline{v}$. Suppose $w = \overline{v}_1$ and $w = \overline{v}_2$. Let $p \in P$ be arbitrary. Then

$$v_1(p) = \overline{v}_1(p) = \overline{v}_2(p) = v_2(p),$$

where the outer equalities hold by the defining property of the bar in Theorem 2.2 ($\overline{v_i}(p) = v_i(p)$), and the middle equality holds since both sides equal w. As p was arbitrary, $v_1 = v_2$. Therefore (d) is true.

(e) If u, w are truth assignments with $u(p) = w(p)$ for all $p \in P$, then $u(\varphi) = w(\varphi)$ for all $\varphi \in \mathrm{Form}(P,S)$.

Response: Both u and w extend the function $v\colon P \to \{\mathrm{T},\mathrm{F}\}$ given by $v(p) = u(p)$ and $v(p) = w(p)$ for all $p \in P$. So by the uniqueness half of Theorem 2.2 $u = w$. So $u(\varphi) = w(\varphi)$ for all $\varphi \in \mathrm{Form}(P,S)$. Therefore (e) is true.

(f) If u is a truth assignment, $w\colon \mathrm{Form}(P,S) \to \{\mathrm{T},\mathrm{F}\}$ is a function, and $u(p) = w(p)$ for all $p \in P$, then w is a truth assignment.

Response: Since u is a truth assignment it extends $v\colon P \to \{\mathrm{T},\mathrm{F}\}$. Suppose $P = \{p\}$. Then let $v(p) = u(p) = w(p) = T$. Let $\phi \in \mathrm{Form}(P,S)$ be have length ≥ 1 with $w(\phi) = T$ Since w is free to disobey the truth tables of the connectives found in S it is a truth assignment. Thus (f) is false.

(g) If u, w are truth assignments with $u(\varphi) = w(\varphi)$ for every formula φ of length at least 1, then $u = w$.

Response: Let $u(\varphi) = w(\varphi)$ for all formulas φ of length ≥ 1, and towards a contradiction assume that there exists a $p \in P$ such that $w(p) \neq u(p)$. WLOG let $u(p) = \mathrm{T}$ and $w(p) = \mathrm{F}$. Since u and w respect the truth table for $\neg$, this forces $u(\neg p) = \mathrm{F}$ and $w(\neg p) = \mathrm{T}$, so $u(\neg p) \neq w(\neg p)$. But $\neg p$ has length 1, so the hypothesis gives $u(\neg p) = w(\neg p)$, a contradiction. Therefore $u(p) = w(p)$ for all $p \in P$. So by (e), $u = w$. Therefore (g) is true.

Problem 2 (Extensions that are not truth assignments)**.** Let $P = \{p\}$, $S = \{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$, and let $v\colon P \to \{\mathrm{T},\mathrm{F}\}$ be given by $v(p) = \mathrm{T}$.

(a) Exhibit two distinct functions $f, g\colon \mathrm{Form}(P,S) \to \{\mathrm{T},\mathrm{F}\}$ with $f(p) = g(p) = \mathrm{T}$, neither of which is a truth assignment. For each, name a specific formula at which it violates a specific truth-table clause, and verify the violation.

Response: Suppose $P = \{p\}$ with $f(p) = g(p) = \mathrm{T}$. For an arbitrary formula ϕ let $f(\phi) = T$. Suppose $f(\neg p) = T$ which violates the truth table for $\neg$. For an arbitrary formula ϕ let $g(\phi) = T$ except for when $\phi = \neg\neg p$. Suppose $g(\neg\neg p) = F$ which also violates the truth table for $\neg$.

(b) Show that there are infinitely many functions $\mathrm{Form}(P, S) \to \{\mathrm{T}, \mathrm{F}\}$ extending v. **Response**:

(c) In light of (b), exactly how many truth assignments extend v? Which result are you citing? **Response**:

(d) Prove that restriction to P and the map $v \mapsto \overline{v}$ are mutually inverse bijections between the set of truth assignments on $\mathrm{Form}(P, S)$ and the set of functions $P \to \{\mathrm{T}, \mathrm{F}\}$. Deduce that if $|P| = n$ then there are exactly 2^n truth assignments on $\mathrm{Form}(P, S)$ (cf. Exercise 2.19). **Response**: